

1. Find the quadratic function that is the best fit for $f(x)$ defined by the table below.

x	0	2	4	6	8	10
f(x)	0	397	1603	3602	6402	9998

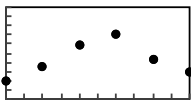
The quadratic function is $y = \boxed{}x^2 + \boxed{}x + (\boxed{})$.
(Type an integer or decimal rounded to two decimal places as needed.)

2. Partially because of people from a certain country living longer, the number with Alzheimer's disease and other dementia is projected to grow each year. The table below gives the millions of citizens in the country with Alzheimer's from 2000 and projected to 2050. Complete parts (a) through (e).

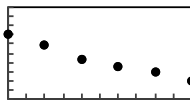
Year	2000	2010	2020	2030	2040	2050
Number	3.9	5.8	6.9	8.7	11.8	14.1

a. Create a scatter plot of the data, with x equal to the number of years after 2000 and y equal to the number of millions of citizens with Alzheimer's disease or other dementia. Choose the correct graph below.

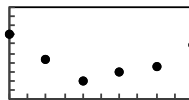
☐ A.



☐ B.



☐ C.



☐ D.



All graphs are shown in $[0, 50]$ by $[0, 20]$ with $X_{\text{scl}}=5$ and $Y_{\text{scl}}=2$ viewing rectangles.

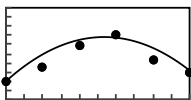
b. Find a quadratic function that models y , the number of millions of people with Alzheimer's disease or other dementia, as a function of x , the number of years after 2000. Report the model with four significant digits.

$y = (\boxed{})x^2 + (\boxed{})x + \boxed{}$
(Type integers or decimals.)

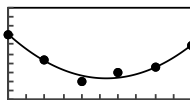
c. Graph the data and the model on the same axes, and comment on the fit.

First graph the data and the model on the same axes. Choose the correct graph below.

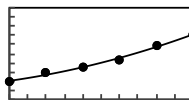
☐ A.



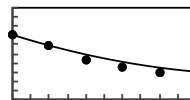
☐ B.



☐ C.



☐ D.



All graphs are shown in $[0, 50]$ by $[0, 20]$ with $X_{\text{scl}}=5$ and $Y_{\text{scl}}=2$ viewing rectangles.

Comment on the fit of the model on the data. Choose the correct answer below.

- ☐ A. The model is a poor fit because it does not pass close to any of the data points.
☐ B. The model is a mediocre fit because it passes close to some of the data points.
☐ C. The model is an excellent fit because it passes through all of the data points.
☐ D. The model is a good fit because it passes close to all of the data points.

d. How many citizens does the model predict will have Alzheimer's disease or other dementia in 2053?

million

(Type an integer or decimal rounded to one decimal place.)

e. During what year after 2000 does the model predict that 9.8 million citizens will have Alzheimer's or other dementia?

3. The table gives the wind-chill temperature when the outside temperature is 20°F. Complete parts (a) through (c) below.

Wind (mph)	Wind Chill (°F)	Wind (mph)	Wind Chill (°F)
5	13	35	0
10	9	40	-1
15	6	45	-2
20	4	50	-3
25	2	55	-3
30	1	60	-4

a. Use x as the wind speed and create a quadratic model for the data.

$$y = (\text{ })x^2 + (\text{ })x + (\text{ })$$

(Type integers or decimals rounded to three decimal places as needed.)

b. At what wind speed does the model estimate that the wind-chill temperature will be -1°F ?

A wind speed of approximately mph will result in a wind chill of -1°F .

(Round to the nearest whole number as needed.)

c. Do you think the model found in part (a) is valid for $x > 58$? Explain. Choose the correct answer below.

- ☐ A. Yes, because the function increases for $x > 58$ and this reflects reality.
- ☐ B. Yes, because the function will continue to decrease for $x > 58$.
- ☐ C. No, because the model is not a good fit.
- ☐ D. No, because the function's minimum is near 57.8 so the model predicts warmer temperatures for winds over 58 mph.